

AFFINE CHANNELS, MIXED-ORDER COEFFICIENT SECTIONS, AND A CONTROLLED UPWARD SHELL RETURN FOR ERDŐS–STRAUS

THE CLANKERS

ABSTRACT. The complete bounded divisor source has an additive fine-defect coordinate. On the exact common-trace locus, the E and M coordinates differ by a fixed affine shift. Their labelled count operator is injective, has an alternating inverse and an orbit-parity cokernel, and reduces partial radix chains to an exact quotient-support cover. We then give a mixed-order subgroup-chain section with a literal inverse, a positive integer coefficient bound, and an optimal finite divisor-DAG test within that certificate class. The section retains cyclic carries and actual exponent multiplicities. Original hard-prime packets with defects 9 and 15 pass this test while the previous unit-direction width test fails. A reduced prime progression realizes an exact missing-quotient boundary. Separately, a labelled factor exchange preserves $4a + s$ while changing the shell and the word; it returns an integral state at the first full shell above the critical $p = 67369$ trace. The same one-shell upward return holds on an explicit infinite progression of hard primes. Neither criterion is asserted to produce an initial source at every prime.

1. PINNED FRONTIER AND PURPOSE

The integration base for this continuation is `KokunoYumeto/erdos-strauss-foundation` commit `6c064a07298cdba38352998364ce6b7cb3efa973`, which incorporates the square-zero/Pell tranche on top of the earlier mixed-fibre and middle-spectrum revision. Its complete source and handoff were read, including the obstruction at $p = 67369$ to repairing every trace without increasing the distinguished denominator. The retained ten-stage plan still requires an original occupied coefficient at every prescribed prime.

We prove a stronger constructive coefficient criterion that uses directions of *different orders*, including nonunits, in the original fine-defect group. It is a cyclic subgroup-transversal construction with exact original multiplicity, carry, and integer-section data. The cyclic calculation is proved from first principles below. No historical-priority claim is made for either the abstract normal form or its arithmetic application. A second construction changes the shell and the word, and gives a one-step upward return at the predecessor's decisive 67369 example. Neither construction assumes that its source exists for every prime.

2. THE ORIGINAL BOUNDED SOURCE AND ITS FINE DEFECT

Fix a prime $p \equiv 1 \pmod{4}$, $p/4 < a < p/2$, and $R = 4a - p$. For every *original* divisor $u \mid a^2$, put

$$a = \prod_i q_i^{e_i}, \quad u = \prod_i q_i^{f_i}, \quad 0 \leq f_i \leq 2e_i, \quad \beta_i = f_i - e_i,$$

$$g = \gcd(a, u), \quad h = g^2/u, \quad r = u/g, \quad s = a/g.$$

Then $a = hrs$, $u = hr^2$, and $\gcd(r, s) = 1$. The ordered rational returns are

$$E : (a, h\kappa, phr\kappa), \quad \kappa = (pr + s)/R, \quad (1)$$

$$M : (a, phs\lambda, phr\lambda), \quad \lambda = (r + s)/R. \quad (2)$$

Direct common-denominator multiplication gives $4/p$ in each case. Their integer gates are $R \mid 4u + 1$ and $R \mid p + 4u$, respectively. The ordered inverses are $u = a^2/(Ry - pa)$ in E and $u = pa^2/(Ry - pa)$ in M. These are the inherited Yamamoto–Rosati/Type-I–II coordinates (Yamamoto, 1965, p. 38, Lemma 2, equation (4))(Elsholtz and Tao, 2013, §2, Propositions 2.2 and 2.6), not a new parametrization. Lopez's structural classification supplies additional literature

context for the difficult class $p \equiv 1 \pmod{24}$ and for Type II solution forms (Lopez, 2024, Introduction and the Type A/B definitions); no theorem of that paper is used to prove the new subgroup-chain or factor-exchange results below.

The rational tail sums are $(a + pu)^2/(Ru)$ and $p(a + u)^2/(Ru)$. Consequently a trace, meaning an integral tail sum, is exactly the square gate $R \mid G^2$, where $G = 4u + 1$ or $G = p + 4u$. Its common tail denominator is $R/\gcd(R, G)$. Define

$$K = \prod_{\ell^v \parallel R} \ell^{\lceil v/2 \rceil}, \quad D = R/K, \quad \delta = R/D^2.$$

Then $R = KD = \delta D^2$, $D \mid K$, and the ideal $K\mathbb{Z}/R\mathbb{Z}$ is square-zero. Here δ is the positive squarefree part, not the radical. For $\eta = 1$ in E and $\eta = 0$ in M, retain the original residue ratio

$$z = p^\eta u/a = p^\eta \prod_i q_i^{\beta_i} \pmod{R}.$$

Trace is exactly $z \equiv -1 \pmod{K}$. Its unique fine coordinate is

$$z = -1 + Kc, \quad c \in \mathbb{Z}/D. \quad (3)$$

The full gate is $c = 0$; the original denominator is $D/\gcd(D, c)$. No integer coefficient is reduced modulo D .

For completeness, partition each original interval $[-e_i, e_i]$ by residue modulo $o_i = \text{ord}_K(q_i)$. Each nonempty part has the literal form

$$\beta_i = b_i + o_i t_i, \quad 0 \leq t_i < N_i.$$

A product part is a trace part when its base is $-1 \pmod{K}$. Put

$$\lambda_i = (q_i^{o_i} - 1)/K \pmod{D},$$

where the division is evaluated modulo R before passing to D . Since $(1 + K\lambda)^t = 1 + tK\lambda \pmod{R}$, its fine coordinate is $c_0 - \sum_i \lambda_i t_i$. Thus the complete original count in that part is

$$C_{c_0}, \quad \mathcal{C} = \prod_i ([0] + [\lambda_i] + \cdots + [(N_i - 1)\lambda_i]) \in \mathbb{Z}[\mathbb{Z}/D]. \quad (4)$$

The inverse reads (b_i, t_i) from β_i and reconstructs $u = \prod_i q_i^{e_i + \beta_i}$. Every vector in the full exponent box belongs to one product part. An explicitly specified subrectangle of such a part also has formula (4); its omitted words remain in the complete source. These source identities are inherited from the supplied square-zero tranche (The Clankers, 2026a, §§2–4); the stronger coefficient criterion follows next.

3. THE EXACT AFFINE RELATION BETWEEN THE TWO CHANNELS

For the same divisor u , put

$$G_E(u) = 4u + 1, \quad G_M(u) = p + 4u.$$

The preceding trace calculation gives

$$E \text{ trace} \iff K \mid G_E(u), \quad M \text{ trace} \iff K \mid G_M(u).$$

Theorem 3.1 (Common traces and the two-channel lattice). *The same original divisor $u \mid a^2$ is a trace in both channels if and only if*

$$K \mid (4u + 1) \quad \text{and} \quad K \mid (p - 1). \quad (5)$$

Equivalently, one channel is a trace and $p \equiv 1 \pmod{K}$. More strongly, if

$$\mathcal{T}_E = \{u \mid a^2 : K \mid 4u + 1\}, \quad \mathcal{T}_M = \{u \mid a^2 : K \mid p + 4u\},$$

then these two trace sets are disjoint when $K \nmid p - 1$ and equal when $K \mid p - 1$.

On the common-trace locus the exact fine coordinates are

$$c_E \equiv \frac{4u + 1}{K} \pmod{D}, \quad c_M \equiv p^{-1} \frac{p + 4u}{K} \pmod{D}. \quad (6)$$

Writing

$$\omega = (p-1)/K, \quad \bar{\omega} = \omega \pmod{D},$$

one has the oriented affine relation

$$c_E = c_M - \bar{\omega} \quad \text{in } \mathbb{Z}/D. \quad (7)$$

Thus M is full exactly at $c_M = 0$, whereas E is full exactly at $c_M = \bar{\omega}$. The same divisor is full in both channels exactly when $R \mid 4u+1$ and $R \mid p-1$.

Let a common trace part have M base coordinate c_0 and coefficient vector C from (4). Its labelled full-state counts are

$$N_M = C_{c_0}, \quad N_E = C_{c_0 - \bar{\omega}}, \quad N_{E \sqcup M} = C_{c_0} + C_{c_0 - \bar{\omega}}. \quad (8)$$

On the full coefficient lattice define

$$(T_{\bar{\omega}} f)_x = f_{x - \bar{\omega}}, \quad J = (I + T_{\bar{\omega}})C.$$

Put $d = \gcd(D, \omega)$ and $L = D/d$. Since D is odd, every $\bar{\omega}$ -orbit has odd length L , and

$$2C_x = \sum_{j=0}^{L-1} (-1)^j J_{x - j\bar{\omega}}. \quad (9)$$

If $\mathcal{O}$ is the set of these d orbits and $\pi(f)_O = \sum_{x \in O} f_x \pmod{2}$, then

$$0 \longrightarrow \mathbb{Z}^{\mathbb{Z}/D} \xrightarrow{I+T_{\bar{\omega}}} \mathbb{Z}^{\mathbb{Z}/D} \xrightarrow{\pi} (\mathbb{Z}/2)^d \longrightarrow 0 \quad (10)$$

is exact. In particular the two-channel operator is injective, and its only integral cokernel is the displayed orbit-parity data.

For $C_x \geq 0$, uniform labelled two-channel positivity is equivalent to

$$\text{supp}(C) \cup (\text{supp}(C) + \bar{\omega}) = \mathbb{Z}/D. \quad (11)$$

It forces the sharp necessary bound

$$|\text{supp}(C)| \geq \frac{D+d}{2}. \quad (12)$$

The adjacency condition (11), not cardinality alone, is the exact criterion.

Proof. The difference $G_M(u) - G_E(u) = p-1$ proves both the common-trace criterion and the trace-set dichotomy. Since $4a \equiv p \pmod{R}$ and a, p are units modulo R ,

$$pu a^{-1} + 1 \equiv 4u + 1 \pmod{R}, \quad u a^{-1} + 1 \equiv p^{-1}(p + 4u) \pmod{R}.$$

Both right sides lie in $K\mathbb{Z}/R$ on their respective trace loci. The map $Kc \mapsto c$ identifies $K\mathbb{Z}/R$ with $\mathbb{Z}/D$, which proves (6) without discarding a factor. On the common locus $p = 1 + K\omega$ and $p \equiv 1 \pmod{D}$, while

$$\frac{p+4u}{K} = \frac{4u+1}{K} + \omega.$$

This proves (7). Replacing divisibility by K with divisibility by $KD = R$ proves the full-gate statements.

For every bounded exponent vector, multiplication of $u/a = -1 + Kc_M$ by $p = 1 + K\omega$ gives

$$p(-1 + Kc_M) = -1 + K(c_M - \omega) + K^2\omega c_M.$$

Because $R \mid K^2$, the last summand vanishes modulo R . The shift is therefore constant over the whole common trace part; its step coefficients and literal exponent intervals are unchanged. Formula (8) follows from the original coefficient definition (4).

On an orbit of $T = T_{\bar{\omega}}$, $T^L = I$ and L is odd. Hence

$$(I + T)(I - T + T^2 - \dots + T^{L-1}) = I + T^L = 2I,$$

which proves (9) and injectivity. Every vector in the image has even coordinate sum on each orbit. Conversely, if those sums are even, every alternating numerator in (9) is even, so the reconstructed C is integral. The parity map is onto by taking one standard basis vector on each orbit. This proves the exact sequence.

Finally, $J_x > 0$ exactly when at least one of $C_x, C_{x-\bar{\omega}}$ is positive, proving (11). Its complement has no two consecutive vertices on any $\bar{\omega}$ -cycle. An independent set in an odd cycle of length L has at most $(L-1)/2$ vertices, so the support has at least $(L+1)/2$ vertices on each of d cycles. This gives (12), and alternating occupancy with one forced adjacent occupied pair on each odd cycle attains the bound. $\square$

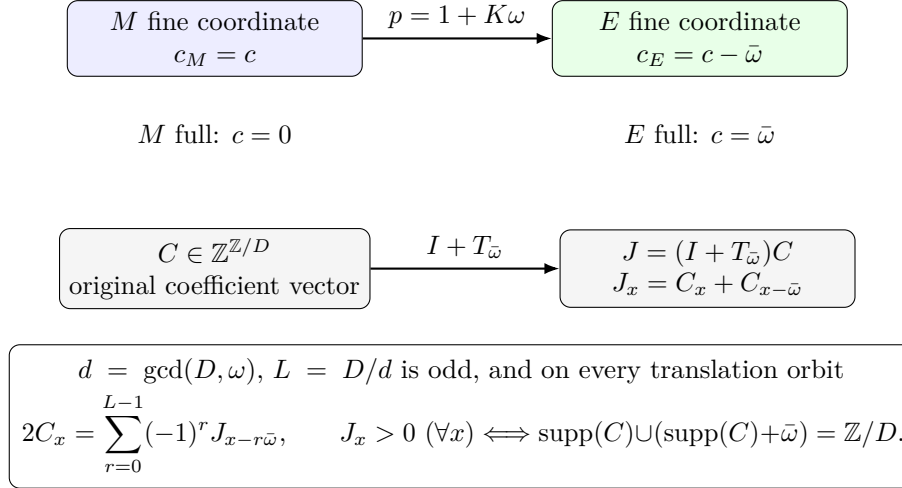


FIGURE 1. The common-trace affine shift, the two labelled full targets, and the exact odd-orbit inverse. The signed inverse shows why adding the two channels does not create positivity from total coefficient mass.

4. MIXED-ORDER SECTIONS WITH ALL INTEGER COEFFICIENTS RETAINED

Let $D \geq 1$, $\lambda_i \in \mathbb{Z}$, and $N_i \geq 1$. Write C_x for the coefficient of $[x]$ in (4). Fix distinct coordinate indices $i_1, \dots, i_k$, and put

$$g_0 = D, \quad g_j = \gcd(g_{j-1}, \lambda_{i_j}), \quad m_j = g_{j-1}/g_j.$$

The nontrivial chain has $g_0 > g_1 > \dots > g_k = 1$.

Theorem 4.1 (Mixed-order coefficient completion). *If $N_{i_j} \geq m_j$ for every j , then every original coefficient satisfies*

$$C_x \geq L := \left(\prod_{i \notin I} N_i \right) \prod_{j=1}^k \left\lfloor \frac{N_{i_j}}{m_j} \right\rfloor > 0. \quad (13)$$

The map

$$(t_{i_1}, \dots, t_{i_k}) \mapsto \sum_j \lambda_{i_j} t_{i_j} \pmod{D}, \quad 0 \leq t_{i_j} < m_j, \quad (14)$$

is a bijection of the $D = \prod_j m_j$ original digit choices with $\mathbb{Z}/D$. It has the explicit reverse algorithm below. When $N_{i_j} = m_j$ for all selected coordinates, every coefficient equals $\prod_{i \notin I} N_i$. The case $D = 1$ is the empty chain, with every word full.

Proof. Let $H_g = g\mathbb{Z}/D\mathbb{Z}$. The quotient $H_{g_j}/H_{g_{j-1}}$ has order m_j . Its specified generator is the image of λ_{i_j} , since $\gcd(g_{j-1}/g_j, \lambda_{i_j}/g_j) = 1$. For $c_k = x \pmod{D}$, compute backwards

$$t_{i_j} \equiv (c_j/g_j)(\lambda_{i_j}/g_j)^{-1} \pmod{m_j}, \quad 0 \leq t_{i_j} < m_j, \quad c_{j-1} = c_j - \lambda_{i_j} t_{i_j} \pmod{D}. \quad (15)$$

At step j , c_j is divisible by g_j ; the chosen digit makes the next residual divisible by g_{j-1} . At the end it is zero modulo D . The same quotient forces each digit of any other representation, proving both existence and uniqueness.

Partition the selected interval $[0, N_{i_j})$ into its $\lfloor N_{i_j}/m_j \rfloor$ full blocks of size m_j , with the remaining short interval separately retained. For every choice of full blocks and every choice of the unselected original coordinates, their offsets translate x . Algorithm (15) gives one unique

original bounded word in those blocks. Distinct choices give distinct words, proving (13). When the selected intervals have exactly the radix lengths, this exhausts all words and proves equality. $\square$

Theorem 4.2 (Partial-chain quotient reduction). *Allow the same strict chain to end at a divisor $g = g_k$ of D , possibly $g > 1$, and retain $N_{i_j} \geq m_j$. Put*

$$B = \prod_{j=1}^k \left\lfloor \frac{N_{i_j}}{m_j} \right\rfloor$$

and, for $y \in \mathbb{Z}/g$, define the exact residual quotient coefficient

$$Q_y = \# \left\{ (t_i)_{i \notin I} : 0 \leq t_i < N_i, \quad \sum_{i \notin I} \lambda_i t_i \equiv y \pmod{g} \right\}.$$

Then there are integers $E_x \geq 0$ such that

$$C_x = BQ_{x \bmod g} + E_x. \quad (16)$$

If $m_j \mid N_{i_j}$ for every selected coordinate, then every $E_x = 0$. For the two common channels of Theorem 3.1, with $\delta = \omega \pmod{g}$,

$$C_x + C_{x-\bar{\omega}} \geq B(Q_{x \bmod g} + Q_{(x \bmod g) - \delta}). \quad (17)$$

Thus the complete-block subsource isolated by this partial chain forces two-channel occupancy for every target precisely when

$$\text{supp}(Q) + \{0, \delta\} = \mathbb{Z}/g. \quad (18)$$

When $g > 1$, write $d_g = \gcd(g, \delta)$. Since g is odd, condition (18) requires the sharp bound

$$|\text{supp}(Q)| \geq (g + d_g)/2. \quad (19)$$

If

$$b = \gcd(g, \{\lambda_i : i \notin I\}) > 1,$$

with $b = g$ when no residual coordinate remains, then condition (18) is impossible.

Proof. For each selected coordinate, retain all complete radix blocks

$$a_j m_j + [0, m_j), \quad 0 \leq a_j < \lfloor N_{i_j}/m_j \rfloor,$$

and retain its terminal interval separately. For fixed block indices the offset is $\sum_j \lambda_{i_j} m_j a_j$. Since

$$\lambda_{i_j} m_j = g_{j-1}(\lambda_{i_j}/g_j),$$

every such offset lies in $g\mathbb{Z}/D$. The bounded radix map on the within-block digits is a bijection onto $g\mathbb{Z}/D$, by the same reverse algorithm (15) stopped at g . Therefore every unselected tuple whose sum equals x modulo g , together with every one of the B block-index tuples, supplies exactly one selected tuple whose total sum is x modulo D . These are $BQ_{x \bmod g}$ distinct original words. All words using at least one terminal interval make up $E_x \geq 0$. There is no terminal interval when every m_j divides N_{i_j} , proving equality in that case.

Apply (16) at x and $x - \bar{\omega}$ to obtain (17). Since all coefficients are nonnegative, its right side is positive at every x exactly under (18). On each δ -orbit in $\mathbb{Z}/g$, the complement of the support can have no adjacent vertices. There are d_g odd cycles of length g/d_g , giving (19) by the odd-cycle argument already used in Theorem 3.1; the same alternating construction attains the bound.

Finally every residual sum in the complete-block subsource lies in the subgroup $b\mathbb{Z}/g$. Its support and one translate occupy at most two cosets of that subgroup. The index is the odd integer $b \geq 3$, so two cosets cannot cover $\mathbb{Z}/g$. This proves the last assertion. In particular, the two channel labels cannot by themselves replace a missing final radix: genuine residual quotient support is necessary. Terminal intervals remain part of the original coefficient vector and may add words outside this certificate. $\square$

Carry, not a direct-product group substitution. The digit map is a set bijection. Its transported group operation is $\text{decode}(\text{encode}(v) + \text{encode}(w))$, not coordinatewise addition in $\prod_j \mathbb{Z}/m_j$. For $D = 9$ and steps $(6, 1)$ with radices $(3, 3)$,

$$(0, 2) \star (0, 1) = (2, 0),$$

because $3 = 2 \cdot 6 \pmod{9}$. The cyclic extension and this integer carry remain.

$$H_9 = 9\mathbb{Z}/9 = \{0\} \xrightarrow{\lambda_1=6, m_1=3} H_3 = 3\mathbb{Z}/9 = \{0, 3, 6\} \xrightarrow{\lambda_2=1, m_2=3} H_1 = \mathbb{Z}/9$$

	$t_1 = 0$	$t_1 = 1$	$t_1 = 2$
$t_6 = 0$	0	1	2
$t_6 = 1$	6	7	8
$t_6 = 2$	3	4	5

$$(0, 2) \star (0, 1) = \text{decode}(2 + 1) = (2, 0)$$

FIGURE 2. The exact $D = 9$ subgroup chain and its carried digit section. The entry in row t_6 and column t_1 is $6t_6 + t_1 \pmod{9}$. Every residue occurs once, but the displayed carry proves that the section is not being replaced by coordinatewise addition in $(\mathbb{Z}/3)^2$.

Fibres and coefficient loss. In the equality case each bin has $\mu = \prod_{i \notin I} N_i$ words. Retaining the unselected digit vector and applying (15) inverts the augmented map from a word to its residue and that retained vector. The residue projection alone has μ -point fibres. Choose one reference word in every residue fibre. On free abelian modules, the differences between each other word and that reference form a basis of the kernel, of rank $D(\mu - 1) = \mu D - D$. All pairwise differences generate the same kernel but are not independent. The coefficient stabilizer is the whole additive group. If μ is zero modulo a prime, none of these original integer words vanishes. Outside the equality case, (13) counts a specified union of full blocks; the remainder and its coefficients stay in (4).

Proposition 4.3 (Optimal bound within the radix-chain class). *There is a finite exact algorithm for the largest lower bound (13) among all chains. Its graph has the divisors $g \mid D$ as vertices, and an edge*

$$g \longrightarrow d = \gcd(g, \lambda_i) < g \quad \text{when} \quad N_i \geq g/d,$$

with score $\lfloor N_i/(g/d) \rfloor / N_i$. Start with score one at D , maximize products along decreasing paths, and multiply the score at one by $\prod_i N_i$. There are at most $n\tau(D)$ candidate edges. If one is unreachable, the zero output means only that this certificate class fails, not that $C_{c_0} = 0$.

Proof. After a coordinate has been used, every subsequent g divides its step. That coordinate can never give another strict edge. Thus paths automatically have distinct labels. Future strict edges depend only on g , not on which prefix reached it. The path score times $\prod_i N_i$ is exactly (13). Dynamic programming on the decreasing divisor graph proves optimality within this stated class. Exact rational scores become integers at every successful terminal path by the displayed product formula. $\square$

The criterion is complementary to, not a replacement for, the earlier unit-width theorem. For $D = 3$, steps $(1, 1)$ and lengths $(2, 2)$ give all three bins, but neither length can start a radix chain. Conversely the examples below pass mixed-order completion while their entire prior unit

width is less than $D - 1$. Even joint generation and a positive source size are insufficient: steps $(3, 1)$ and lengths $(2, 2)$ generate $\mathbb{Z}/9$ but miss five bins.

If an original full coefficient is absent, every feasible radix chain on its trace part must therefore stop before one. The set of reachable divisors, and the failed inequality $N_i < g/\gcd(g, \lambda_i)$ at each missing outgoing edge, are an exact additional obstruction certificate. This does not assert that such certificates are contradictory across all shells.

5. ACTUAL PRIME ARITHMETIC WITH GENUINELY MIXED ORDERS

The following are original bounded packets, not freely chosen residue supports. Set $R = 243$, $K = 27$, $D = 9$ and

$$a = H \cdot 53 \cdot 163 \cdot 271, \quad p = 4a - 243, \quad H \equiv 29 \pmod{210}.$$

For every prime member $p > R$ retain the M packet

$$\beta_{53} \in \{-1, 1\}, \quad \beta_{163}, \beta_{271} \in \{-1, 0, 1\},$$

with every other original centred exponent zero. If H contains one of these primes, the actual larger exponent interval is still retained and this is its specified subpacket. Modulo K the first prime is -1 and the other two are one. The packet polynomial, up to its retained base shift, is

$$(1 + X^5)(1 + X^6 + X^{12})(1 + X + X^2) = 2 \sum_{c=0}^8 X^c \pmod{X^9 - 1}.$$

The chain is $9 \rightarrow 3 \rightarrow 1$ using steps 6, 1, both with radix three. The preceding unit width is only $1 + 2 = 3 < 8$. One full word is

$$u = H(163 \cdot 271)^2, \quad (h, r, s, \lambda) = (H, 44173, 53, 182),$$

since $44173 + 53 = 243 \cdot 182$. Its original complement is the other full word.

There is an all-nonunit selected-chain example. Put

$$R = 675, \quad K = 45, \quad D = 15, \quad a = H \cdot 89 \cdot 271^2 \cdot 1801, \quad p = 4a - 675, \quad H \equiv 101 \pmod{210}.$$

The original M packet has centred ranges $\{-1, 1\}$ at 89, $[-2, 2]$ at 271, and $[-1, 1]$ at 1801, with the other coordinates zero. Its polynomial is

$$(1 + X^{11}) \left(\sum_{j=0}^4 X^{6j} \right) (1 + X^{10} + X^{20}) = 2 \sum_{c=0}^{14} X^c \pmod{X^{15} - 1}.$$

The selected steps 6, 10 are both nonunits; their chain is $15 \rightarrow 3 \rightarrow 1$ with radices 5, 3. The only unit width is the single binary 89 coordinate, so the older unit-width criterion does not apply. One full word and its marking are

$$u = H \cdot 1801^2, \quad (h, r, s, \lambda) = (H, 1801, 6536249, 9686).$$

Both examples are uniform original coefficient vectors with every bin equal to two. Freezing either selected mixed direction at centred exponent zero removes the full words of the specified M packet; the executable tables check these exact restricted fibres as well.

These are infinite prime-source statements, not an assumption of prime values of a nonlinear polynomial. With $A = \prod q_i^{e_i}$ fixed, $p = 4AH - R$ and $H = H_0 + 210n$ give a linear progression of modulus $840A$. It is reduced: $p \equiv 1 \pmod{840}$ and $p \equiv -R \pmod{q_i}$ for each fixed prime q_i , which is a unit. Dirichlet's theorem therefore supplies infinitely many prime members (Sutherland, 2017, Theorem 18.1). Each such member has the constructed source and full M return. This does not cover arbitrary prescribed primes. The JSON tables select particular prime members by a bounded deterministic search and certify their primality independently by complete-order certificates.

5.1. Arithmetic sharpness at an arbitrary odd defect. The last radix capacity cannot be lowered even on actual hard-prime packets. Fix odd $D > 1$ and integers $\lambda_1, \dots, \lambda_k$ such that

$$g_0 = D, \quad g_j = \gcd(g_{j-1}, \lambda_j), \quad D = g_0 > g_1 > \dots > g_k = 1.$$

Put $m_j = g_{j-1}/g_j$, let the last radix be $m = m_k = g_{k-1} > 1$, and put $R = 3D^2$, $K = 3D$. For $i < k$, choose distinct primes greater than seven with

$$q_i \equiv 1 + K\lambda_i \pmod{R}.$$

Choose a further prime

$$q_k \equiv -1 + Kc \pmod{R}, \quad 2c \equiv -\lambda_k \pmod{D}.$$

Every displayed class is a unit modulo R , so Dirichlet supplies these primes. For $i < k$ one has $\text{ord}_K(q_i) = 1$ and $(q_i - 1)/K \equiv \lambda_i \pmod{D}$. Since $K^2 = 3R$,

$$q_k^2 = (-1 + Kc)^2 \equiv 1 - 2Kc \pmod{R},$$

so $\text{ord}_K(q_k) = 2$ and

$$\frac{q_k^2 - 1}{K} \equiv -2c \equiv \lambda_k \pmod{D}.$$

Take original exponents $e_i = (m_i - 1)/2$ for $i < k$ and $e_k = m - 1$. Let $A = \prod q_i^{e_i}$, and impose

$$p \equiv 1 \pmod{840}, \quad p \equiv 4A - R \pmod{4A^2}.$$

The congruences agree modulo four and are reduced at every other prime. Infinitely many prime members therefore have $a = (p + R)/4 = AH$ with $H \equiv 1 \pmod{A}$, so the stated prime exponents are exact.

Hold the other centred coordinates zero. For $i < k$ use all $\beta_i \in [-e_i, e_i]$; for q_k use the odd part

$$\beta_k \in \{-(m-2), -(m-4), \dots, m-2\}.$$

For $i < k$, square-zero expansion gives $q_i^{\beta_i} \equiv 1 + K\lambda_i\beta_i \pmod{R}$; because β_k is odd, $q_k^{\beta_k} \equiv -1 + Kc\beta_k \pmod{R}$. Hence every displayed word is an original M trace and its original residue ratio and fine defect are

$$z = \prod_i q_i^{\beta_i} \equiv -1 + K \left(c\beta_k - \sum_{i < k} \lambda_i \beta_i \right) \pmod{R}, \quad d(\beta) = c\beta_k - \sum_{i < k} \lambda_i \beta_i \pmod{D}.$$

For $i < k$, the centred interval $[-e_i, e_i]$ has exactly m_i members and is a complete digit interval modulo m_i . The reverse decoder therefore maps their product bijectively onto $H_m = m\mathbb{Z}/D$, whose order is $D/m = \prod_{i < k} m_i$. Moreover $\gcd(c, m) = 1$, because $2c \equiv -\lambda_k \pmod{m}$, m is odd, and $\gcd(\lambda_k, m) = 1$. The $m - 1$ displayed values of β_k are exactly the nonzero residue classes modulo m , so their multiples by c are the nonzero quotient classes of $(\mathbb{Z}/D)/H_m$. Thus the actual fine-defect count $\mu_x = \#\{u : d(u) = x\}$ is

$$\mu_x = \begin{cases} 0, & m \mid x, \\ 1, & m \nmid x. \end{cases} \quad (20)$$

Its actual coefficient stabilizer is exactly H_m : a translation preserves the zero set precisely when it lies in H_m . The source maps bijectively onto the complement of H_m . In particular zero is absent, despite all preceding capacities being exact. This construction does not furnish or claim an ES counterexample: it proves an exact empty subpacket while retaining every other original word.

The separate finite realization has

$$D = 9, \quad K = 27, \quad R = 243, \quad (\lambda_1, \lambda_2) = (6, 1), \quad (q_1, q_2) = (163, 107),$$

$$A = 163 \cdot 107^2 = 1866187, \quad H = 523, \quad a = 976015801, \quad p = 3904062961.$$

Its six centred exponent pairs $\beta_{163} \in \{-1, 0, 1\}$, $\beta_{107} \in \{-1, 1\}$ have fine-defect vector

$$(\mu_0, \dots, \mu_8) = (0, 1, 1, 0, 1, 1, 0, 1, 1), \quad \text{Stab}(\mu) = \{0, 3, 6\} = 3\mathbb{Z}/9.$$

This $H = 523$ example is distinct from the displayed $H \equiv 1 \pmod{A}$ lift. The reduced exact-exponent progression through it is

$$p = 3904062961 + 2925429291933960t.$$

Complete enumeration finds no full word in the displayed $R = 243$ shell. The same prime nevertheless has the exterior full witness

$$a_0 = 976015741, \quad R_0 = 3, \quad u_0 = 89, \quad (h, r, s, \kappa) = (89, 1, 10966469, 1305009810),$$

and therefore the exact E denominators

$$(976015741, 1273710116719419210, 453440801203675619490).$$

Thus the boundary packet is not an ES counterexample. Complete prime, word, progression, stabilizer and exterior-witness certificates are supplied.

6. A FACTOR EXCHANGE THAT PERMITS CONTROLLED UPWARD MOVEMENT

The supplied least-example theorem shows that a repair from *every* trace cannot always satisfy $a' \leq a$ (The Clankers, 2026a, §7). We retain another exact integer invariant instead. For any original shell and any chosen divisor $s \mid a$, write $n = a/s$ and define

$$C = R + s, \quad M = p + C = 4a + s = (4n + 1)s. \quad (21)$$

Here $0 < C < 3p/2$ and $M < 5p/2$.

Theorem 6.1 (Complete labelled plus-factor fibre). *At fixed p, C , every positive first-half integer cell with invariant (21) is exactly*

$$\begin{aligned} s' \mid M = p + C, \quad M/s' &\equiv 1 \pmod{4}, \quad 0 < C - s' < p, \\ n' = (M/s' - 1)/4, \quad R' = C - s', \quad a' = n's' &= (M - s')/4. \end{aligned} \quad (22)$$

The map is inverted by retaining s and s' . Its unit-ray exterior targets are exactly those cells additionally satisfying $R' \mid M$. For each,

$$(h', r', s', \kappa') = (n', 1, s', M/R' - 1), \quad u' = n', \quad (23)$$

with ordered denominators $(a', n's'\kappa', pn'\kappa')$. The change of first denominator is exactly

$$a' - a = (s - s')/4.$$

Proof. Equation (21) proves both directions of (22); its congruence implies $R' \equiv -p \equiv 3 \pmod{4}$, so the strict range gives $R' \geq 3$ and $p/4 < a' < p/2$. Because $s' \mid a'$ and $a' < p$, one has $\gcd(p, s') = 1$; moreover

$$\gcd(R', s') = \gcd(4n's' - p, s') = \gcd(p, s') = 1.$$

Since $M = (4n' + 1)s'$, the original unit-ray E gate $R' \mid 4n' + 1$ is therefore equivalent to $R' \mid M$. Equivalently, $p + s' = M - R'$, so $R' \mid p + s'$ precisely when $R' \mid M$. Then (23) is the inherited full E identity, and $u' = n'$ divides a'^2 with its actual original exponents. Conversely a full unit-ray target in this fibre forces that same divisibility. Both residual and carrier factors are computed, not declared favourable. $\square$

The whole cell list has at most $\tau(M)$ members. The full target list is a specified finite divisor test, not an assertion that every fibre has a target. For a canonical original E or M trace, choose s from its exact gcd factorization. Then $n = hr$. For a marked unit-ray E target, its complete *canonical* incoming trace fibre is: enumerate every cell (22), every $r \mid n$ with $\gcd(r, s) = 1$, put

$$h = n/r, \quad u = nr,$$

and test each original square gate. This reconstructs all fields and all original word bounds. An M word already retains its order. On ordered rational solution triples define the exact E-tail involution

$$\tau_E(a, Y, Z) = (a, Z, Y), \quad \tau_E^2 = \text{id}.$$

This permutation does not generally preserve the original E exponent box, so it is retained as a map of ordered solutions and is not inserted as another member of the canonical incoming word fibre. In particular it is not an extra free bit on an already counted M word.

The labelled arrow includes its chosen target. Forgetting that choice makes the forward relation set-valued when more than one divisor passes. On free integer coefficient modules, choose one reference arrow over every target. The differences between each other arrow and its fibre reference form a kernel basis; all pairwise same-target differences generate that kernel but are dependent. No group-ring reduction changes the count of distinct original source words or of distinct target states.

6.1. The critical 67369 source moves up by one shell. At

$$p = 67369, \quad a = 16849 = 7 \cdot 29 \cdot 83, \quad R = 27, \quad u = 4067,$$

the exact M gcd factors are $(h, r, s) = (83, 7, 29)$ and $\lambda = 4/3$. Thus $C = 56$, $M = 67425$. Select

$$s' = 25, \quad R' = 31, \quad n' = 674, \quad a' = 16850.$$

Both 25 and 31 divide 67425, with $67425/25 \equiv 1 \pmod{4}$. The resulting marking is

$$(a', R', u', h', r', s', \kappa') = (16850, 31, 674, 674, 1, 25, 2174),$$

and

$$\frac{4}{67369} = \frac{1}{16850} + \frac{1}{36631900} + \frac{1}{98714178844}. \quad (24)$$

The old factors 7, 29, 83 and the new factors 2, 5, 337 do not share a prime. The invariant is $4a + s = 4a' + s'$, not an identification of their factor boxes.

$$p = 67369, \quad C = 56, \quad M = p + C = 67425$$

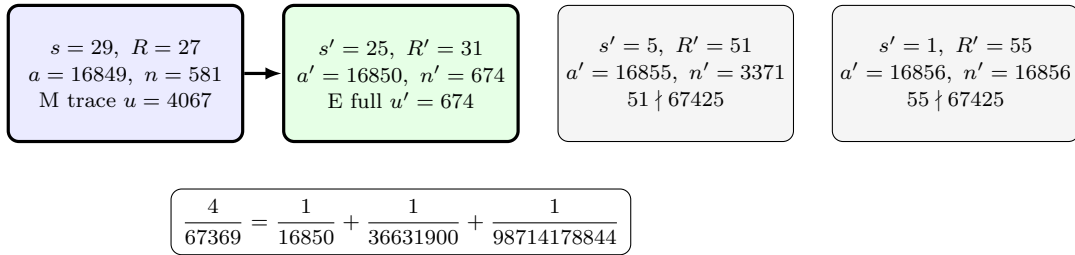


FIGURE 3. The complete $p = 67369$, $C = 56$, $M = 67425$ plus-factor cell set. The arrow retains the source word and selected target. Only the $s' = 25$ cell passes the unit-ray terminal test $R' \mid M$; the other divisor cells remain in the fibre rather than being discarded.

The release enumerates every divisor at residuals 3, 7, 11, 15, 19, 23, 27, 31. The first seven have no full E/M word; $R = 27$ has proper traces and $R = 31$ has full E words 674, 84250. This rechecks the relevant fixed-prime obstruction; it does not claim to have newly replayed the predecessor's leastness proof over all 817 smaller primes. The chosen target's canonical incoming trace fibre is computed by the complete rule above, including any full source itself.

The return extends to a reduced prime progression, not just one example. Put

$$h = 83 + 4650k, \quad p = 812h - 27 = 67369 + 3775800k, \quad k \geq 0.$$

Every prime member is 169 (mod 840) and has the proper source

$$(a, R, u, h, r, s, \lambda) = (203h, 27, 49h, h, 7, 29, 4/3).$$

The target is

$$a' = 203h + 1, \quad R' = 31, \quad u' = h' = (203h + 1)/25, \quad r' = 1, \quad s' = 25, \quad \kappa' = (p + 25)/31.$$

All divisions are exact. The progression is reduced, so infinitely many prime members exist. The old word and ray deletion systems fail on this source: their moduli exceed 27, while the

properness requires deletion by a multiple of three. The earlier factor-sum map to $\delta = 3$ also fails because its necessary divisibility is $7 \mid (3^2 - 1)/4 = 2$. The new formula always has $a' = a + 1$. Only the initial prime is asserted to have an entirely empty old full shell; no such extra assertion is made for every prime in the progression.

6.2. Why the new fibre is not automatically occupied. At $p = 97, a = 40, u = 2$ in M , the canonical s is 20, so $C = 83$ and $M = 180$. The complete cell list is

$$(a', R', s') = (36, 47, 36), (40, 63, 20), (44, 79, 4).$$

None has a unit-ray E target. Exhausting each of these three complete original square-divisor boxes finds no full E or M state at all. The input is nevertheless a proper M trace. This refutes universal occupancy of a fixed plus-factor fibre, not ES or possible transformations between different such fibres.

7. REMAINING ARITHMETIC OBLIGATION AND VERIFICATION SCOPE

The mixed-order theorem forces original target coefficients when its actual quotient capacities pass; its inverse and all integer multiplicities are known. The upward return crosses a previously certified obstruction to every nonincreasing repair at its selected source. The two mechanisms are not implicitly composable on every input: any application must preserve the shown source, interval, channel and divisibility tests.

The affine-channel theorem now gives an exact alternative to a full radix chain. A partial chain may stop at $g > 1$ only if its surviving quotient support and its δ -translate cover $\mathbb{Z}/g$. Two channel labels do not constitute the missing final radix: when the residual steps remain in a proper subgroup, the support meets at most two of at least three quotient cosets. Thus a proof through the two-channel route must construct the required quotient support, or introduce a proved source-changing map that alters this obstruction. Total word mass, a norm bound, or the injectivity of $I + T_{\bar{\omega}}$ cannot replace that step.

If ES failed at a prescribed prime, every full coefficient in the complete first-half source would be zero. Each trace part would fail both the previous unit-width test and every feasible mixed-order chain. Every chosen plus-factor fibre reachable by a proposed argument would have to miss its displayed terminal divisor test. These are exact necessary conditions, not a contradiction by definition. The remaining Stage 7 step is still to force at least one original gate, with controlled changes between shells if required.

The standard-library replay exhausts the complete original boxes at primes $p \equiv 1 \pmod{24}$ through its recorded bound and checks every squareful trace part against literal gates. It additionally examines 161887 finite additive polynomials, compares the divisor-DAG optimum with an independent ordered-subsequence implementation, retains all nonzero carries, and verifies complete original example boxes and inverse fibres. The separate checker imports neither the main program nor predecessor code. Large example primes have full order certificates whose factors are themselves certified; trial division is used in discovery and as a separate finite check. Finite search caps, outcomes, and file hashes are in the release receipt. A probable-prime label is not used as a proof. There is no Lean build, independent human review, priority conclusion, or new overall ES verification-range claim.

7.1. Primality and finite-check boundary. The isolated large primes have complete-order certificates. A nonterminal node retains the full certified prime factorization of $n - 1$ and a base b with

$$b^{n-1} \equiv 1 \pmod{n}, \quad \gcd(b^{(n-1)/q} - 1, n) = 1 \quad (q \mid n - 1).$$

For any prime divisor ℓ of n , the order of b modulo ℓ divides $n - 1$ and does not divide $(n - 1)/q$ for any prime q of $n - 1$. Thus that order is $n - 1$, which divides $\ell - 1$. It follows that $\ell \geq n$, proving n prime. Terminal nodes below 10000 are checked by complete trial division through their integer square root. Discovery tests and completed certificates are recorded separately. The arithmetic census tests every source word at its finite bound; it is not an additional assertion about primes beyond that bound.

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